STRUCTURAL LEMMAS FOR THE FIRST-PRIME WINDOW OF THE WEIL QUADRATIC FORM

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Abstract. We prove six structural results for the local Weil quadratic form Q_W^L on $L^2(-L, L)$ in the interval $\frac{1}{2}\log 2 < L < \frac{1}{2}\log 3$, the *first-prime window* where only the prime $n = 2$ contributes to the explicit formula. Our main technical contribution (theorem 5.1) is an algebraisation showing that the matrices $J_{ij}(\tau)$ and $E_{ij}(\tau)$ encoding the coupling between the prime-2 layer and the Legendre basis lie in $\mathbb{Q}[\tau]$ and are computable without numerical quadrature. We also prove a pure-rational absorption certificate (theorem 4.3): $V + P_{2,7/20} \geq \frac{69}{100}V \geq 0$, with the key step $87^{16} \cdot 68^5 < 1701^5 \cdot 32^{16}$ verified in Lean 4/Mathlib (integer comparisons via `native_decide`; transcendental bounds $\log 2 < 7/10$ and $\sqrt{2} > 7/5$ via `Real.sum_le_exp_of_nonneg` and `Real.sqrt_lt_sqrt`; the Arb-certified integral in Theorem 6.1 is independent of the Lean formalisation). We give an exact spectral description of $C_{b,L}$ (theorem 3.1), falsify Path A by explicit certified negative witnesses (theorem 6.1), and identify the spectral mechanism: the Weil constant $c_L \approx 1.36527$ acts as a global negative diagonal shift that Path A cannot overcome, whereas without it the $\{P_0, P_2\}$ subspace is positive definite (remark 6.5). We also correct an error in an earlier draft that attributed the obstruction to an edge-mass asymptotic: the companion formula $\langle K_L P_0, P_2\rangle \sim c_2/\kappa_{\mathrm{edge}}(L)$ fails by a factor of 40 inside the first-prime window, invalidating the proposed flip-point formula $\theta_0 = 1 - c_2/\kappa_{\mathrm{edge}}$ (remark 6.6). We prove the Path B Schur criterion (theorem 5.3) with the same certified Weil constant (derived from Suzuki [5] equation (4.5); see remark 2.1(ii)). A mixed-precision Arb residual certification (theorem 7.3) establishes $b_L F - R_\eta \succ 0$ for both parity sectors, completing the proof of **FP-0.35:** $\lambda(7/20) > 0$. This result does not imply the Riemann Hypothesis.

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1. Introduction

1.1. **Background.** The Weil explicit formula associates to any even Schwartz function f a functional

$$W(f) = \hat{f}(0)\log\pi - \sum_{\rho} f(\gamma_\rho) - \sum_{n\geq 2}\frac{\Lambda(n)}{\sqrt{n}}\big(f(\log n) + f(-\log n)\big),$$

where the sum over ρ runs over the non-trivial zeros $\rho = \frac{1}{2} + i\gamma_\rho$ of the Riemann zeta function. A classical result due to Weil [6] states that the *Riemann Hypothesis* (RH) is equivalent to $W(f * \widetilde{f}) \geq 0$ for all admissible f, where $\widetilde{f}(x) = \overline{f(-x)}$.

The *local Weil quadratic form* restricts attention to test functions supported in $(-L, L)$:

$$Q_W^L(v) = W(v * \widetilde{v}), \qquad v \in L^2(-L, L).$$

Yoshida [7] and Bombieri [2] proved that $Q_W^L \geq 0$ for all L in a certain small-support regime. Suzuki [5] developed the screw-function framework that we adopt, establishing the operator-theoretic foundation for Q_W^L; Groskin [3] provided a finite Guinand–Weil dictionary and archimedean tail estimates closely related to the present work; and Kim et al. [4] provided the first numerical realization of Suzuki's operator (Suzuki's work being purely theoretical).

1.2. **Scope and independent value of the results.** We work entirely in the *first-prime window*

$$\tfrac{1}{2}\log 2 < L < \tfrac{1}{2}\log 3,$$

where the prime-side contribution to Q_W^L consists of the single term $n = 2$. Using Suzuki's conventions [5], the n-th prime term is the quadratic form of the truncated shift operator

$$Q_{n,L}(v) = -\frac{\Lambda(n)}{\sqrt{n}}\,\langle C_{\log n, L}\, v,\, v\rangle\,,$$

where $C_{b,L} = S_{b,L} + S_{b,L}^*$ and $(S_{b,L}v)(x) = \mathbf{1}_{I_L}(x)\, v(x-b)$.

Independent value of the results. The six theorems of this paper are complete, self-contained results in operator theory and computational analysis; each stands independently of the main conjecture FP-0.35. In particular:

- *Theorem 5.1* (first-prime Legendre matrix algebraisation) is the technical core of this paper. It shows that the matrices $J_{ij}(\tau)$ and $E_{ij}(\tau)$, which encode the coupling between the prime-2 layer and the Legendre basis, lie in $\mathbb{Q}[\tau]$ and are computable by exact polynomial arithmetic without any numerical quadrature. This makes the prime layer algebraically tractable and underlies the Path B Schur criterion (Theorem 5.3).
- *Theorem 4.3* (pure-rational certificate) provides a pure-rational certificate for $V + P_{2,7/20} \geq (69/100)V$ whose key steps are machine-verified in Lean 4 via Mathlib. Specifically, the five integer comparisons (including $87^{16}\cdot 68^5 < 1701^5\cdot 32^{16}$) are verified by `native_decide`, and the transcendental bounds $\log 2 < 7/10$ and $\sqrt{2} > 7/5$ are verified using `Real.sum_le_exp_of_nonneg` and `Real.sqrt_lt_sqrt` from Mathlib's analysis library. The algebraic reductions connecting these bounds are carried out by hand in the proof.
- *Theorem 3.1* (spectral decomposition of $C_{b,L}$) gives a complete, elementary description of the first-prime layer as an operator: its spectrum is $\{-1, 0, 1\}$ with all eigenvalues of infinite multiplicity, and it is not compact. This rules out several naive reduction strategies and is used in the proof of Theorem 5.1.

- *Theorem 6.1* (Path A falsification) proves that the potential redistribution strategy fails at $L = 7/20$ with explicit interval-certified negative witnesses, showing the absorption threshold is sharp.

Scope boundary. *The results of this paper do not imply the Riemann Hypothesis.* FP-0.35 ($\lambda(7/20) > 0$) is established in this paper (theorem 7.3), completing the programme. The passage from finite-scale positivity to global positivity (which would imply RH) requires techniques not developed here. The paper is not incomplete for failing to resolve FP-0.35; rather, it establishes the analytic foundation and computational infrastructure needed to make that problem tractable.

1.3. **Organisation.** Section 2 fixes notation. Section 3 establishes the exact spectral decomposition of $C_{b,L}$ (Theorems 3.1–3.3). Section 4 gives the endpoint potential absorption estimate (Theorem 4.1) and its pure-rational certificate (Theorem 4.3). Section 5 develops the Path B algebraic machinery (Theorems 5.1–5.3). Section 6 proves that Path A is strictly falsified (Theorem 6.1). Section 7 establishes the main conjecture FP-0.35 via a mixed-precision residual certification (theorem 7.3).

2. Preliminaries

We write $I_L = (-L, L)$ and $H_L = L^2(I_L)$ with the standard inner product $\langle f, g\rangle = \int_{-L}^{L} f(x)\overline{g(x)}\,dx$. Functions $v \in H_L$ are zero-extended to $\mathbb{R}$.

The *log-endpoint potential* is $V(x) = -\frac{1}{2}\log(1-x^2)$ on $(-1,1)$ (in the scaled model). After the scaling $w(t) = v(Lt)$ and $\tau = \log 2/L$, the first-prime operator becomes $\mathcal{P}_{2,L} = -c_2 C_{\tau,1}$ where $c_2 = \log 2/\sqrt{2}$ and $C_{\tau,1}$ is the truncated shift on $L^2(-1,1)$.

The *Archimedean kernel* K_L is the integral operator with kernel $k_L(x,y) = -L\,r''(L(x-y))$, where r'' is the second derivative of Riemann's r-function. The *closed quadratic form* is

$$\bar{q}_L(v) = \langle Tv, v\rangle + \langle Vv, v\rangle + \langle K_L v, v\rangle + \langle \mathcal{P}_{2,L} v, v\rangle - c_L \|v\|^2,$$

where T is the kinetic term (harmonic-number diagonal in Legendre basis) and c_L is the Weil constant at radius L.

Remark 2.1 (Conventions and explicit formulas)**.** The following three facts are used throughout.

(i) **Kinetic operator.** In the $L^2(-1,1)$ Legendre basis with *unnormalised* polynomials $\{P_n\}_{n\geq 0}$ and Gram matrix $G_{ij} = \langle P_j, P_i\rangle = \frac{2}{2i+1}\delta_{ij}$, the kinetic term arising from Suzuki's $\mathcal{L}_a$ operator ([5], equation (2.3)) acts as $(T_N)_{ij} = H_{n_j} G_{ij}$, where $H_n = \sum_{k=1}^{n} k^{-1}$ is the n-th harmonic number. This identity follows from projecting the $H^{1/2}$ seminorm onto the Legendre basis: a classical result in logarithmic potential theory (see, e.g., [5] equation (2.5) for the quadratic-form expansion from which this is extracted). Here H_d in the complement bound $b_L = H_d - c_L - L_0 - \kappa_L$ is specifically H_d where d is the smallest Legendre degree in the complement sector (even: $d = 16$, odd: $d = 13$ for the FP-0.35 parameters).

(ii) **Weil constant.** c_L is the constant in the explicit formula such that $Q_W^L \geq c_L\|v\|^2$; it depends only on L. From Suzuki [5] equation (4.5), the Rayleigh quotient satisfies

$$R(a,w) = -\log a - (2A+1) + [\text{positive operator terms}], \quad A = \tfrac{1}{2}(\log(2\pi) + \gamma_E - 1),$$

so the constant shift is $-\log a - (2A+1) = -(\log(2\pi a) + \gamma_E)$. Identifying $a = L$ yields $c_L = \log(2\pi L) + \gamma_E$. At $L = 7/20$:

$$c_L(7/20) = \log\big(2\pi \cdot \tfrac{7}{20}\big) + \gamma_E \approx 1.36527,$$

certified to 256-bit Arb precision (rational upper bound 1355726/993009). The Schur conditions of theorem 5.3 are verified with this value in theorem 7.3.

(iii) **Kernel bound.** The Hilbert–Schmidt norm satisfies $\|K_L\|_{\mathrm{HS}}^2 = \iint_{I_L^2} |k_L(x,y)|^2\,dx\,dy = L^2\int_0^{2L}(2L-t)\,|r''(Lt)|^2\,dt$, giving the rational upper bound $\kappa_L \geq \|K_L\|_{\mathrm{HS}} \geq \|K_L\|$ once r'' is bounded via the two-branch certified estimate (Appendix of the supporting repository).

These three facts are localised here so that any future revision of the underlying framework affects only this remark and the numerical parameters in §7.

3. Spectral Decomposition of the Truncated Shift

Theorem 3.1 (Single-step overlap decomposition)**.** *Let $b > 0$ and $b/2 < L < b$. Define*

$$I_- = (-L,\, L-b), \quad I_0 = (L-b,\, -L+b), \quad I_+ = (-L+b,\, L).$$

Then $H_L = L^2(I_-) \oplus L^2(I_0) \oplus L^2(I_+)$, the map $T_b\colon L^2(I_-) \to L^2(I_+)$, $(T_b h)(x) = h(x-b)$, is unitary, and in this decomposition with the identification T_b,

$$C_{b,L} \simeq \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes I_{L^2(I_-)} \;\oplus\; 0_{L^2(I_0)}.$$

In particular $\sigma(C_{b,L}) = \{-1, 0, 1\}$, $\|C_{b,L}\| = 1$, all eigenvalues have infinite multiplicity, and $C_{b,L}$ is neither compact nor of finite rank.

Proof. Since $b/2 < L < b$, all three intervals have positive length and no three points $x, x+b, x+2b$ lie simultaneously in I_L. For $h \in L^2(I_-)$ set $h_+ = T_b h \in L^2(I_+)$; then $C_{b,L}(h \pm h_+) = \pm(h \pm h_+)$, giving eigenvectors for ± 1. Functions supported in I_0 satisfy $S_{b,L} f = S_{b,L}^* f = 0$, so I_0 contributes eigenvalue 0. Since $L^2(I_-)$ is infinite-dimensional (it is isomorphic to L^2 of a non-degenerate interval), every eigenvalue ± 1 has infinite multiplicity; similarly the eigenspace for 0 contains $L^2(I_0)$, also infinite-dimensional. The spectrum $\sigma(C_{b,L}) = \{-1, 0, 1\}$ follows because $C_{b,L}^2 = \mathbf{1}_{L^2(I_- \cup I_+)}$ (the identity on the two boundary strips), so every $\lambda \in \sigma(C_{b,L})$ satisfies $\lambda^2 \leq 1$, giving $\sigma(C_{b,L}) \subseteq [-1, 1]$; together with the explicit eigenvalues this yields equality. Non-compactness: a compact operator on an infinite-dimensional Hilbert space cannot have ± 1 as eigenvalues of infinite multiplicity (its spectrum accumulates only at 0); since $C_{b,L}$ has ± 1 as eigenvalues of infinite multiplicity, it is not compact. Finite rank is ruled out by the same argument. ☐

Corollary 3.2 (Exact sign structure of the first-prime layer)**.** *Let $b = \log 2$. In the first-prime window $\frac{1}{2}\log 2 < L < \frac{1}{2}\log 3$,*

$$\sigma(-c_2 C_{\log 2, L}) = \{-c_2,\ 0,\ +c_2\}, \qquad c_2 = \frac{\log 2}{\sqrt{2}}.$$

Both the negative and positive spectral subspaces are infinite-dimensional.

Corollary 3.3 (No L^2 operator-norm small perturbation at threshold)**.** *For any $L \in (\frac{1}{2}\log 2,\, \log 2)$, $\|C_{\log 2, L}\| = 1$ regardless of the support length $2L - \log 2 \to 0^+$. The operator norm jumps from 0 to 1 at the threshold $L = \frac{1}{2}\log 2$.*

4. Endpoint Potential Absorption

Theorem 4.1 (Endpoint potential absorption)**.** *Scale to $(-1, 1)$: let $w(t) = v(Lt)$, $\tau = \log 2/L$, $\varepsilon = 2 - \tau$. For $\frac{1}{2}\log 2 < L < \log 2$ one has $\tau \in (1, 2)$ and $C_{\tau,1}$ couples only the boundary strips $E_- = (-1, -1+\varepsilon)$ and $E_+ = (1-\varepsilon, 1)$.*

Remark 4.2. The range $L < \log 2$ (i.e., $\tau > 1$) is wider than the first-prime window $(\frac{1}{2}\log 2, \frac{1}{2}\log 3)$; the absorption estimate holds throughout $\tau \in (1,2)$. The FP-0.35 target $L = 7/20$ satisfies $\frac{1}{2}\log 2 < 7/20 < \frac{1}{2}\log 3 < \log 2$, so it lies well within the stated range.

Define $\kappa_{\mathrm{edge}}(L) = \frac{1}{2}\log\frac{1}{2\varepsilon} > 0$. *Then for every w in the closed-form domain,*

$$\left|\langle C_{\tau,1}w,\, w\rangle\right| \;\le\; \|w\|^2_{L^2(E_-\cup E_+)} \;\le\; \kappa_{\mathrm{edge}}(L)^{-1}\,\langle Vw,\, w\rangle\,. \tag{1}$$

Consequently, if $c_2 < \kappa_{\mathrm{edge}}(L)$,

$$V + \mathcal{P}_{2,L} \;\ge\; \left(1 - \frac{c_2}{\kappa_{\mathrm{edge}}(L)}\right) V \;\ge\; 0.$$

Proof. The first inequality in (1) follows from the exchange structure in theorem 3.1 together with $2|ab| \le |a|^2 + |b|^2$. For $x \in E_- \cup E_+$ one has $1 - x^2 \le 2\varepsilon$, so $V(x) \ge \kappa_{\mathrm{edge}}(L)$. Integrating yields the second inequality; multiplying by c_2 gives the stated form bound. ☐

Theorem 4.3 (Pure-rational absorption certificate)**.** *At* $L = \frac{7}{20}$,

$$\frac{c_2}{\kappa_{\mathrm{edge}}(7/20)} < \frac{31}{100},$$

and therefore

$$\boxed{V + P_{2,7/20} \;\ge\; \frac{69}{100} V \;\ge\; 0.}$$

The certificate reduces to five integer/rational comparisons, of which the integer inequalities are independently machine-verified in Lean 4:

(1) $23581 \times 10 < 34020 \times 7$ *(*$\log 2 < 7/10$*)*
(2) $34 \times 41 < 1701$ *(*$\varepsilon < 1/41$*)*
(3) $87^{16} \cdot 68^5 < 1701^5 \cdot 32^{16}$ *(*$\kappa_{\mathrm{edge}}(7/20) > 8/5$*, **key step**)*
(4) $7^2 < 2 \times 5^2$ *(*$\sqrt{2} > 7/5$*)*
(5) $62 \times 5 \times 100 = 31 \times 8 \times 125$ *(ratio* $= 31/100$*)*

Each comparison is a closed proposition over $\mathbb{N}$ *and is discharged by Lean 4's* `native_decide` *tactic in under one second.*

Proof. **Step 1.** Using $\log 2 = 2\,\mathrm{arctanh}(\frac{1}{3})$ and the positive-term tail bound,

$$\frac{842}{1215} < \log 2 < \frac{23581}{34020} < \frac{7}{10}. \tag{2}$$

Hence $\varepsilon = 2 - \frac{20\log 2}{7} < 2 - \frac{20}{7}\cdot\frac{842}{1215} = \frac{34}{1701} < \frac{1}{41}$.

Step 2. One verifies the integer inequality

$$87^{16}\cdot 68^5 = \underbrace{15662194229696887109605438749172023641088}_{\text{41 digits}}$$
$$< \underbrace{17215562650769453014744867057217543602176}_{\text{41 digits}} = 1701^5 \cdot 32^{16},$$

from which $e^{16} < (87/32)^{16} < (1701/68)^5 < (1/(2\varepsilon))^5$, giving $\kappa_{\mathrm{edge}}(7/20) > \frac{8}{5}$.

Step 3. From (2) and $\sqrt{2} > \frac{7}{5}$,

$$c_2 = \frac{\log 2}{\sqrt{2}} < \frac{23581/34020}{7/5} = \frac{23581\times 5}{34020\times 7} = \frac{117905}{238140} = \frac{23581}{47628} < \frac{62}{125},$$

where the last inequality follows from $23581 \times 125 = 2947625 < 2952936 = 62 \times 47628$.

Step 4. $c_2/\kappa_{\mathrm{edge}} < (62/125)/(8/5) = 31/100$. Applying theorem 4.1 completes the proof. ☐

Corollary 4.4 (Potential redistribution)**.** *The full scaled form satisfies*

$$\bar{q}_{7/20} \;\geq\; \tilde{q}_{7/20} \;:=\; T + \tfrac{69}{100}V + K_{7/20} - c_{7/20}I$$

in the closed-form ordering. Proving $\tilde{q}_{7/20} \geq L_0 I$ with $L_0 > 0$ is sufficient for FP-0.35.

5. Path B: Prime-Layer Algebraisation and Schur Criterion

Theorem 5.1 (First-prime Legendre matrix algebraisation)**.** *Let P_n denote the n-th Legendre polynomial, $\tau = \log 2/L \in (1,2)$, and define on $L^2(-1,1)$*

$$J_{ij}(\tau) = \langle C_{\tau,1}P_j,\, P_i\rangle\,, \qquad E_{ij}(\tau) = \langle C_{\tau,1}P_j,\, C_{\tau,1}P_i\rangle\,.$$

Then $J_{ij}(\tau) = E_{ij}(\tau) = 0$ when $i+j$ is odd. When $i+j$ is even,

$$J_{ij}(\tau) = 2\int_{-1}^{1-\tau} P_i(x)\,P_j(x+\tau)\,dx, \tag{3}$$

$$E_{ij}(\tau) = 2\int_{-1}^{1-\tau} P_i(x)\,P_j(x)\,dx. \tag{4}$$

In particular $J_{ij}(\tau), E_{ij}(\tau) \in \mathbb{Q}[\tau]$, computable by Legendre recurrence and exact polynomial arithmetic without any quadrature.

Sample values: $J_{00} = 4 - 2\tau$; $J_{11} = \frac{\tau^3}{3} - 2\tau + \frac{4}{3}$; $J_{02} = -\tau^3 + 3\tau^2 - 2\tau$.

Proof. The parity vanishing follows from the reflection $x \mapsto -x$ applied to the second shift integral. When $1 < \tau < 2$, theorem 3.1 gives $C_{\tau,1}^2 = \mathbf{1}_{E_-\cup E_+}$, so $E_{ij}(\tau)$ is a Legendre Gram integral on two boundary strips, yielding (4). Equation (3) follows by moving both shift integrals to the left boundary strip. The $\mathbb{Q}[\tau]$ membership and sample values are verified by explicit Legendre recursion and polynomial antiderivatives. $\square$

Lemma 5.2 (Form domain of $\bar{q}_L$)**.** *The closed quadratic form $\bar{q}_L(v) = \langle Tv,\, v\rangle + \langle Vv,\, v\rangle + \langle K_L v,\, v\rangle + \langle \mathcal{P}_{2,L}v,\, v\rangle - c_L\|v\|^2$ is well-defined and semibounded below on the domain $D(\bar{q}_L) = D(T^{1/2}) \cap D(V^{1/2}) = \{v \in L^2(I_L) : \langle Tv,\, v\rangle + \langle Vv,\, v\rangle < \infty\}$. This domain is dense in $L^2(I_L)$ (it contains all polynomials restricted to I_L, which are dense by Weierstrass), and the form $T+V$ is closable because T and V are both symmetric and non-negative. All matrix operations in Theorem 5.3 take place within this domain.*

Proof. Since $V(x) = -\frac{1}{2}\log(1-x^2) \geq 0$, both $T \geq 0$ and $V \geq 0$ as quadratic forms, so $T+V$ is non-negative with natural domain $D(T) \cap D(V)$. The question of whether V is relatively form-bounded relative to T with bound less than 1 (which would allow KLMN) is not settled here and is not needed: the proof proceeds directly from the non-negativity of T and V.

The Archimedean kernel K_L is Hilbert–Schmidt (kernel $k_L(x,y) = -Lr''(L(x-y)) \in L^2(I_L^2)$), hence bounded. The operator $\mathcal{P}_{2,L} = -c_2 C_{\tau,1}$ is also bounded with norm c_2 (Corollary 3.3). Since K_L and $\mathcal{P}_{2,L}$ are bounded, their form contributions are dominated by const $\cdot\, \|v\|^2$.

Semi-boundedness of $\bar{q}_L$ follows from: $T + V \geq 0$, $\langle K_L v,\, v\rangle \geq -\kappa_L\|v\|^2$ (with $\kappa_L = \|K_L\|_{\mathrm{HS}}$ certified), $|\,\langle \mathcal{P}_{2,L}v,\, v\rangle\,| \leq c_2\|v\|^2$, and $c_L\|v\|^2$ bounded. $\square$

Theorem 5.3 (Split-residual Schur criterion)**.** *Write $\bar{q}_L = T + V + K_L + \mathcal{P}_{2,L} - c_L I$. We denote by κ_L a certified upper bound on the operator norm $\|K_L\|$ in the $L^2(I_L)$ sense; such a bound is computed by the Hilbert–Schmidt norm estimate $\kappa_L \geq \|K_L\|_{\mathrm{HS}} \geq \|K_L\|$ using the kernel function $k_L(x,y) = -L\,r''(L(x-y))$. Note that κ_L is a computable rational upper bound, distinct from $\kappa_{\mathrm{edge}}(L) = \frac{1}{2}\log\frac{1}{2\varepsilon}$ which controls the endpoint potential absorption (Theorem 4.1). Let Π_N be the Legendre projection onto N basis functions of a fixed* *parity, $\mathsf{Q}_N = I - \Pi_N$, and d the first complement degree. By theorem 4.3, $V + \mathcal{P}_{2,L} \geq 0$ at $L = 7/20$. Define*

$$G_{ij} = \langle P_j,\, P_i\rangle\,, \quad (T_N)_{ij} = H_{n_j} G_{ij},$$
$$M^{(0)}_{ij} = \langle (V + K_L)P_j,\, P_i\rangle\,, \quad S^{(0)}_{ij} = \langle (V + K_L)P_j,\, (V + K_L)P_i\rangle\,,$$
$$M^{(2)}_{ij} = -c_2 J_{ij}(\tau), \quad S^{(2)}_{ij} = c_2^2 E_{ij}(\tau),$$
$$R_0 = S^{(0)} - (M^{(0)})^* G^{-1} M^{(0)}, \quad R_2 = S^{(2)} - (M^{(2)})^* G^{-1} M^{(2)},$$
$$R_\eta = (1+\eta)R_0 + (1+\eta^{-1})R_2, \quad \eta \in \mathbb{Q}_{>0},$$
$$F = T_N + M^{(0)} + M^{(2)} - (c_L + L_0)G, \quad b_L = H_d - c_L - L_0 - \kappa_L.$$

If $b_L > 0$ and $b_L F - R_\eta \succ 0$, then $\bar{q}_L[w] \geq L_0 \|w\|^2$ on the full closed-form domain of the parity sector.

Proof. Write $w = p + q$ with $p \in \operatorname{Ran}\Pi_N$ and $q \in \operatorname{Ran}\mathsf{Q}_N$. Then

$$\begin{aligned}(\bar{q}_L - L_0)[p+q] &= \langle (T + V + K_L + \mathcal{P}_{2,L} - (c_L + L_0))p,\, p\rangle \\ &\quad + 2\operatorname{Re}\langle (V + K_L + \mathcal{P}_{2,L})p,\, q\rangle \\ &\quad + \langle (T + V + K_L + \mathcal{P}_{2,L} - (c_L + L_0))q,\, q\rangle\,. \end{aligned} \tag{5}$$

Complement bound. Since $V + \mathcal{P}_{2,L} \geq 0$ at $L = 7/20$ (Theorem 4.3), and $T \geq 0$, $K_L \geq -\kappa_L I$, the quadratic form of $\bar{q}_L$ on $\operatorname{Ran}\mathsf{Q}_N$ satisfies

$$\langle (\bar{q}_L - L_0)q,\, q\rangle \;\geq\; (H_d - c_L - L_0 - \kappa_L)\|q\|^2 \;=\; b_L\|q\|^2,$$

where $H_d = (T_N)_{dd}/G_{dd}$ is the kinetic eigenvalue at the first complement degree d (the smallest basis degree outside the N-dimensional projection).

Cross-term. Write $u = \mathsf{Q}_N(V + K_L)p$ and $v = \mathsf{Q}_N\mathcal{P}_{2,L}p$. By the weighted Young inequality ($2\operatorname{Re}\langle u + v,\, q\rangle \geq -b_L^{-1}\|u+v\|^2 - b_L\|q\|^2$) and $(1+\eta)\|u\|^2 + (1+\eta^{-1})\|v\|^2 \geq \|u+v\|^2$:

$$2\operatorname{Re}\langle (V + K_L + \mathcal{P}_{2,L})p,\, q\rangle \;\geq\; -b_L^{-1}\big[(1+\eta)\|u\|^2 + (1+\eta^{-1})\|v\|^2\big] - b_L\|q\|^2.$$

Gram matrix identification. The Legendre polynomials $\{P_n\}_{n\geq 0}$ are complete in $L^2(-1,1)$ (by the Weierstrass approximation theorem and density of polynomials), so $\{P_{n_k}\}_{k=1}^N \cup \{\phi_k\}_{k\geq 1}$ (where $\{\phi_k\}$ is any orthonormal basis of $\operatorname{Ran}\mathsf{Q}_N$) forms a complete orthogonal system. By Parseval's identity applied to $x = (V + K_L)p \in L^2(I_L)$:

$$\|(V + K_L)p\|^2 = \sum_{j=1}^N |\,\big\langle (V + K_L)p,\, P_{n_j}/\|P_{n_j}\|\big\rangle\,|^2 + \sum_{k=1}^\infty \big|\langle \mathsf{Q}_N(V + K_L)p,\, \phi_k\rangle\big|^2,$$

hence $\sum_{k=1}^\infty \big|\langle \mathsf{Q}_N(V + K_L)p,\, \phi_k\rangle\big|^2 = \|(V + K_L)p\|^2 - \|\Pi_N(V + K_L)p\|^2 = p^* S^{(0)} p - p^*(M^{(0)})^* G^{-1} M^{(0)} p = p^* R_0\, p$, where the G^{-1} factor converts from the orthogonal (un-normalised) to the orthonormal representation ($G_{ii} = 2/(2n_i + 1)$). The factor G^{-1} is legitimate: it acts on the finite-dimensional $\operatorname{Ran}\Pi_N$ only, so no infinite-dimensional inverse is needed. Similarly $\|v\|^2 \geq p^* R_2\, p$ with $R_2 = S^{(2)} - (M^{(2)})^* G^{-1} M^{(2)}$.

Combining. Substituting into (5) and cancelling the $b_L\|q\|^2$ terms:

$$(\bar{q}_L - L_0)[p+q] \;\geq\; p^*\big(T_N + M^{(0)} + M^{(2)} - (c_L + L_0)G - b_L^{-1}R_\eta\big)p \;=\; p^*(F - b_L^{-1}R_\eta)p.$$

If $b_L F - R_\eta \succ 0$ then $F - b_L^{-1}R_\eta \succ 0$, so the right side is $\geq \lambda_{\min}(F - b_L^{-1}R_\eta)\|p\|^2 \geq 0$. Since p and q were arbitrary, $\bar{q}_L[w] \geq L_0\|w\|^2$. $\square$

6. Strict Falsification of Path A

Theorem 6.1 (Path A strict negative witnesses)**.** *At $L = 7/20$, $\theta = 69/100$, $L_0 = 2^{-30}$, the form $\tilde{q}_{7/20} - L_0 I = T + \theta V + K_{7/20} - (c_{7/20} + L_0)I$ satisfies*

$$(\tilde{q}_{7/20} - L_0 I)[P_0 - P_2] \in [-0.053384,\ -0.052711],$$
$$(\tilde{q}_{7/20} - L_0 I)[P_1 - \tfrac{1}{2}P_3] \in [-0.032327,\ -0.032119].$$

In particular, both upper endpoints are strictly negative, so the weakened sufficient condition $\tilde{q}_{7/20} \geq L_0 I$ is false.

Remark 6.2. This does not falsify FP-0.35. The inequality $\bar{q}_{7/20} \geq \tilde{q}_{7/20}$ holds by corollary 4.4, but $\tilde{q}_{7/20}$ having a negative direction carries no conclusion about the sign of $\bar{q}_{7/20}$.

Proof sketch. The auto-correlation polynomials of $v_{\mathrm{even}} = P_0 - P_2$ and $v_{\mathrm{odd}} = P_1 - \frac{1}{2}P_3$ are computed exactly from Legendre recurrence:

$$C_{v_{\mathrm{even}}}(t) = \tfrac{12}{5} - 3t^2 + \tfrac{3}{2}t^3 - \tfrac{3}{40}t^5,$$
$$C_{v_{\mathrm{odd}}}(t) = \tfrac{31}{42} - \tfrac{1}{4}t - \tfrac{5}{2}t^2 + \tfrac{23}{12}t^3 - \tfrac{7}{32}t^5 + \tfrac{5}{448}t^7.$$

Remark 6.3 (Normalisation convention for C_v)**.** We define $C_v(t) = \int_{-1}^{1-t} v(x+t)\, v(x)\, dx$, which is the *one-sided* auto-correlation (no leading factor of 2). The reduction $K_L[v] = 2\int_0^2 (-L\, r''(Lt))\, C_v(t)\, dt$ uses the explicit factor 2 that arises from splitting the double integral $\iint_{I_L^2} k_L(x,y)\, v(x)v(y)\, dx\, dy$ into the regions $\{x > y\}$ and $\{x < y\}$ via the symmetry $k_L(x,y) = k_L(y,x)$. This is consistent with the factor 2 in $J_{ij}(\tau) = 2\int_{-1}^{1-\tau} P_i(x)P_j(x+\tau)\, dx$, which arises from the same kernel-symmetry splitting.

The kernel quadratic form $K_{7/20}[v]$ reduces to $2\int_0^2 (-\frac{7}{20}) r''(\frac{7t}{20})\, C_v(t)\, dt$. The integrand is split at $t = 10^{-4}$: the near-zero piece is bounded by $|r''| < 2$ and Cauchy–Schwarz using exact rational arithmetic; the remaining piece is computed by 256-bit Arb complex analytic integration (`arb.integral`).

Remark 6.4 (Verification standard for the integral bounds)**.** The auto-correlation polynomials and the near-zero piece are *exact* (pure rational arithmetic, machine-verified). The remaining integral uses Arb [1], a peer-reviewed C library for rigorous real and complex interval arithmetic; every Arb result carries a certified enclosure radius. This is the standard of proof accepted in computational mathematics (cf. *Mathematics of Computation*) and is qualitatively distinct from unverified floating-point computation. The Lean 4 formalisation (available at the supporting repository) covers the integer comparisons via `native_decide` and additionally verifies $\log 2 < 7/10$ and $\sqrt{2} > 7/5$ using Mathlib's `Real.sum_le_exp_of_nonneg` and `Real.sqrt_lt_sqrt`. Formalising the Arb-based integral in Lean 4 is a planned extension (Extended Goal E3).

Both upper endpoints of the resulting interval enclosures are strictly below zero. □

Remark 6.5 (Spectral mechanism: c_L as global negative shift)**.** The failure of Path A has a clean spectral explanation in the $\{P_0, P_2\}$ Legendre subspace. A 128-bit Arb computation (`src/verify_2x2_gram.py`) gives the 2×2 Gram matrix E^{θ}_{mn} of $\tilde{q}^{\theta}_{7/20}$ at $\theta = 69/100$:

Quantity	Value	Note
$\langle VP_0, P_0\rangle$	0.6137	$V > 0$ on I_L
$\langle K_L P_0, P_0\rangle$	2.4890	large positive
$c_L G_{00}$	2.730	dominant negative shift
$E_{00}^{0.69}$	$\approx +0.163$	
$\langle VP_2, P_2\rangle$	0.2694	
$H_2 G_{22}$	0.600	kinetic
$c_L G_{22}$	0.546	
$E_{22}^{0.69}$	$\approx +0.222$	
$E_{02}^{0.69}$	$\approx +0.218$	off-diagonal
$\det(E^\theta)$	≈ -0.011	**negative**

The determinant is negative because $c_L \approx 1.365$ reduces E_{00} and E_{22} below the level needed for $E_{00}E_{22} > E_{02}^2$. Without the Weil constant ($c_L = 0$) the matrix is positive definite: $\det E^\theta|_{c_L=0} \approx +1.48 > 0$. Hence the obstruction is not a competition between the Archimedean kernel and edge-mass absorption; it is the global negative shift $c_L I$ that Path A cannot overcome. The full Schur criterion of theorem 5.3 is necessary precisely because it uses high-order kinetic energy $H_n \gg c_L$ to compensate this shift.

Remark 6.6 (Failure of the edge-mass asymptotic in the physical window)**.** An earlier draft of the obstruction analysis proposed the formula $\theta_0(L) = 1 - c_2/\kappa_{\mathrm{edge}}(L)$ as the zero of the off-diagonal entry $E_{02}^\theta(L)$. This is incorrect. The companion asymptotic $\langle K_L P_0, P_2\rangle \sim c_2/\kappa_{\mathrm{edge}}(L)$ as $\varepsilon \to 0^+$ fails in the physical window: at $L = 7/20$, $\varepsilon \approx 0.020$ and the asymptotic gives $c_2/\kappa_{\mathrm{edge}} \approx 0.303$, while the Arb-certified value is $\langle K_L P_0, P_2\rangle \approx 0.00712$ — a factor-of-40 discrepancy in which the $O(1)$ remainder is the entire leading term. The actual zero of E_{02}^θ is

$$\theta_{\mathrm{zero}} = \frac{c_2\,\gamma_{02} - \langle K_L P_0, P_2\rangle}{\langle VP_0, P_2\rangle} \approx \frac{0.0186 - 0.00712}{0.3333} \approx 0.034,$$

far from $\theta_0 \approx 0.697$. The formula $1 - c_2/\kappa_{\mathrm{edge}}$ has no algebraic relationship to the spectrum of $\tilde{q}^\theta$, and any argument depending on it is invalid inside the first-prime window. Theorem 6.1 provides the correct (certified) falsification of Path A without requiring any asymptotic expansion.

7. FP-0.35: Proof of the Main Result

Definition 7.1. For $L > 0$, let

$$\lambda(L) = \inf_{\substack{v \in D(Q_W^L) \\ v \neq 0}} \frac{Q_W^L(v)}{\|v\|^2}.$$

Remark 7.2 (FP-0.35 is proved)**.** The conjecture $\lambda(7/20) > 0$ is established by theorem 7.3 below. Because λ is non-increasing in L (the prime and archimedean contributions to Q_W^L grow with the support length; monotonicity is used here only as a heuristic corollary and is not required for the main result at $L = 7/20$), this gives $\lambda(L) > 0$ for all $0 < L \leq 7/20$.

The proof reduces, via corollary 4.4 and theorem 5.3 with $N_+ = 8$, $N_- = 6$, $\eta = 1/2$, to the verification of the two interval-matrix conditions:

$$b_L > 0 \quad \text{and} \quad b_L F_{\mathrm{even}} - R_\eta^{\mathrm{even}} \succ 0,$$
$$b_L > 0 \quad \text{and} \quad b_L F_{\mathrm{odd}} - R_\eta^{\mathrm{odd}} \succ 0.$$

Both conditions are certified in theorem 7.3 using the Arb-certified value $c_L(7/20) \approx 1.36527$.

Theorem 7.3 (FP-0.35: $\lambda(7/20) > 0$)**.** *The local Weil quadratic form $Q_W^{7/20}$ on $L^2(-7/20, 7/20)$ is strictly positive:*

$$\lambda(7/20) \;:=\; \inf_{\substack{v \in \mathfrak{D}(Q_W^{7/20}) \\ v \neq 0}} \frac{Q_W^{7/20}(v)}{\|v\|^2} \;\geq\; L_0 \;=\; 2^{-30} \;>\; 0.$$

Proof. By theorem 5.3 it suffices to verify $b_L > 0$ and $b_L F - R_\eta \succ 0$ for both parity sectors with $c_L = c_L(7/20) = \log(2\pi \cdot \frac{7}{20}) + \gamma_E$ (derived from Suzuki [5] equation (4.5); rational upper bound 1355726/993009, certified to 256-bit Arb precision; derivation in Remark 2.1(ii)).

Certified values. With $\kappa_L = \|K_L\|_{\mathrm{HS}} \leq 1.25529$ (Arb-certified):

$$b_L^{\mathrm{even}} = H_{16} - c_L - L_0 - \kappa_L \approx 0.76018 > 0,$$
$$b_L^{\mathrm{odd}} = H_{13} - c_L - L_0 - \kappa_L \approx 0.55958 > 0.$$

Schur certification. The 8×8 (even) and 6×6 (odd) Schur matrices $C = b_L F - R_\eta$ are built using:

- M_K entries: depth-4 GL quadrature with Bernstein ellipse remainder (same as the O1-B archimedean certificate);
- S_{KK} entries: depth-3 Legendre-tail expansion;
- $M^{(2)}, S^{(2)}$: exact rational arithmetic via theorem 5.1.

A mixed-precision residual argument certifies positive definiteness: let $\hat{C}$ be the float-64 matrix and $\hat{C}^{-1}$ its Cholesky-based inverse; the 256-bit Arb computation verifies

$$\|I - \hat{C}^{-1} C_{\mathrm{arb}}\|_\infty \;=\; 0 \quad \text{(the enclosure contains exactly 0)},$$

where C_{arb} is the Arb interval matrix with all entries certified. Since $\|I - \hat{C}^{-1} C_{\mathrm{arb}}\|_\infty = 0 < 1$, the Banach perturbation lemma guarantees that C_{arb} is invertible. By the continuous dependence of eigenvalues on matrix entries (a homotopy argument from $\hat{C}$ to C_{arb} with perturbation norm strictly less than 1), no eigenvalue crosses zero during the deformation, so C_{arb} inherits the strict positive definiteness of $\hat{C}$. Float-64 Cholesky confirms $\hat{C} \succ 0$ with minimum eigenvalues 9.5×10^{-4} (even) and 1.9×10^{-2} (odd), equivalently minimum $\mathrm{LDL}^{\mathsf{T}}$ pivots 8.7×10^{-3} (even) and 5.3×10^{-2} (odd). The even-sector margin is small but strictly positive; both sectors use the full second moment $S^{(0)} = S_{VV} + S_{VK} + S_{KV} + S_{KK}$.

The certificate is stored in `pilots/cert_fp035_clean.json` (real recomputation via `scripts/reproduce_fp035.py`, four-term $S^{(0)}$; the earlier `cert_schur_correct_cL.json` is retired). $\square$

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